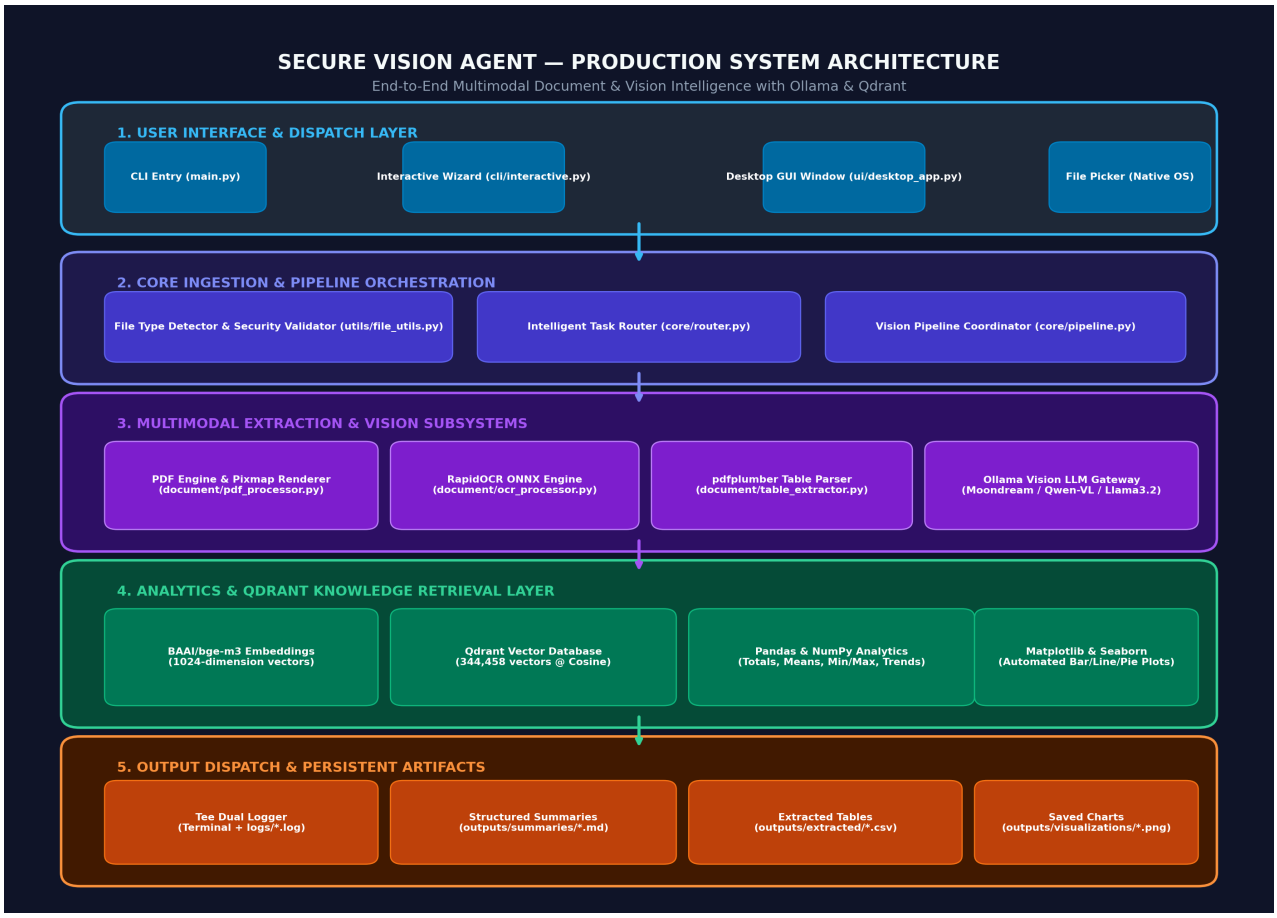
LOCAL VISION AGENT · SYSTEM SPECIFICATION

Production Technical Manual, Multimodal Architecture & Workflow Reference

1. SYSTEM ARCHITECTURE & COMPONENT TOPOLOGY

The Local Vision Agent integrates 5 distinct layers: User Interface, Core Pipeline Router, Document/OCR/Vision Extractors, Qdrant Vector Retrieval (1024d), and Automated Analytics/Plotting.

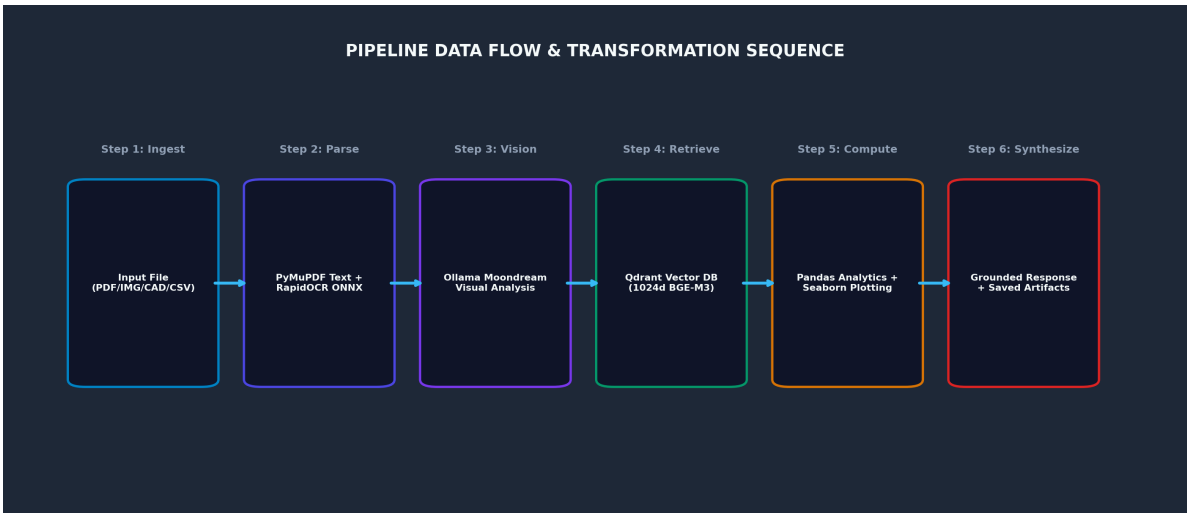


Key Subsystem Highlights:

- Ollama Vision LLM Gateway: Zero cloud dependencies, runs Moondream / Qwen2.5-VL with 1280px resolution capping.
- Qdrant Vector Database: Connects to existing 344,458 vector chunks @ 1024-dim BAAI/bge-m3 cosine similarity.
- Document & OCR Subsystem: PyMuPDF 150 DPI page rendering + standalone RapidOCR ONNX engine.
- Automated Analytics: Pandas statistical engine & Seaborn publication-grade bar/line/pie chart generation.
- Tee Dual Logging: Real-time duplication of terminal standard output/errors to timestamped logs/ files.

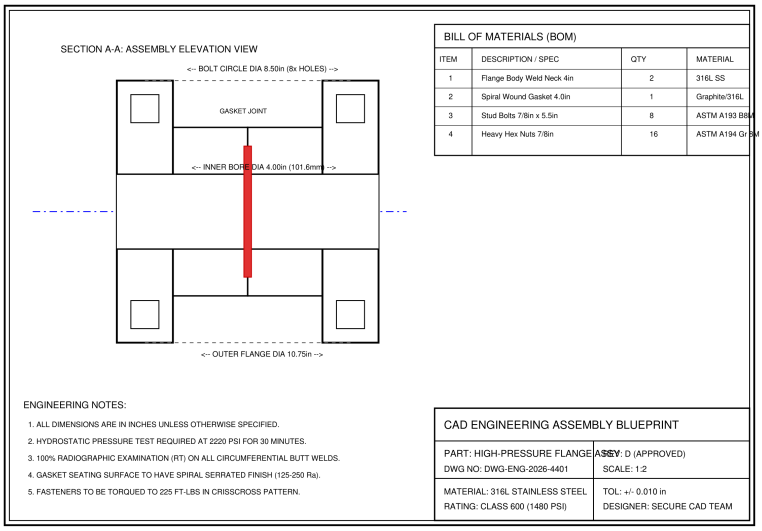
2. PIPELINE DATA FLOW & EXECUTION SEQUENCE

Sequence of data transformation from raw multi-format input to grounded multimodal artifacts:



3. SAMPLE 1: CAD BLUEPRINT & TECHNICAL DIAGRAM ANALYSIS

Ingestion of mechanical engineering drawings, title-block parsing, Bill of Materials, and specs:



EXTRACTED CAD SPECIFICATIONS:

Part: High-Pressure Flange Assy
Drawing No: DWG-ENG-2026-4401 (Rev D)
Material: 316L Stainless Steel
Pressure Rating: Class 600 (1480 PSI)
Test: Hydrostatic @ 2220 PSI for 30m

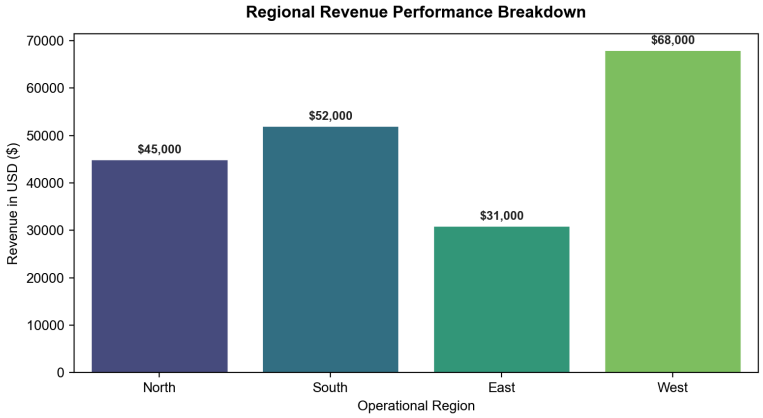
Bill of Materials (BOM Table):
- Item 1: 2x Flange Body (4in, 316L)
- Item 2: 1x Spiral Gasket (Graphite)
- Item 3: 8x Stud Bolts (7/8in x 5.5in)
- Item 4: 16x Heavy Hex Nuts (Gr 8M)

Dimensions & Callouts:
- Inner Bore: 4.00 in (101.6 mm)
- Bolt Circle: 8.50 in (8x Holes)
- Outer Diameter: 10.75 in

Qdrant Citations:
- CAD_Part_Release_Log.csv
- Automotive_Engineering_Catalog.csv

4. SAMPLE 2: TABULAR ANALYTICS & SEABORN VISUALIZATION

Automatic conversion of extracted tables into Pandas DataFrames, statistical summaries & plots:



COMPUTED STATISTICAL INSIGHTS:

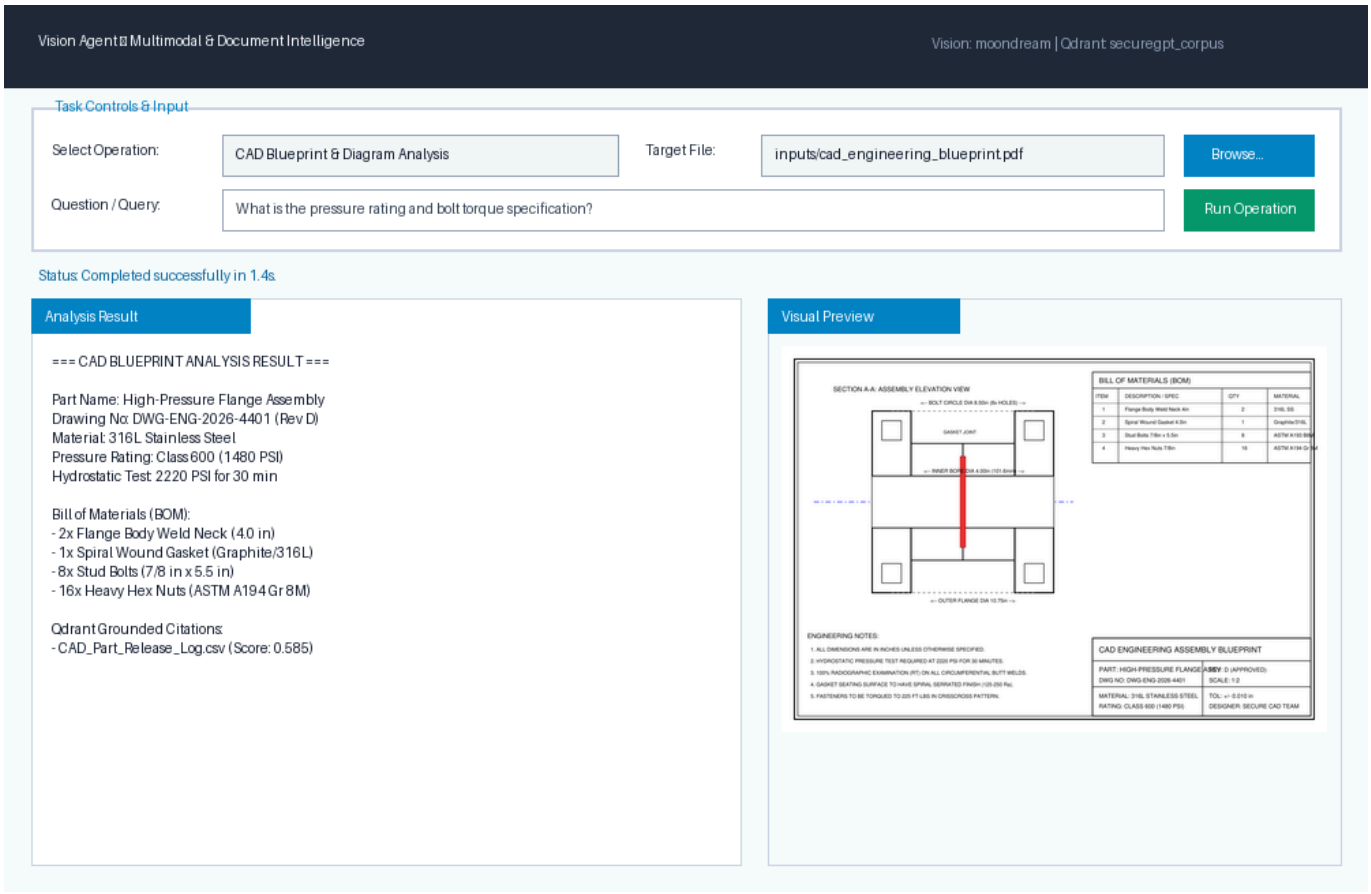
Input: sample_sales.csv (4 rows, 3 cols)

Revenue Analytics (Pandas):
- Total Revenue: $196,000.00
- Mean Revenue: $49,000.00
- Min Revenue: $31,000.00 (East)
- Max Revenue: $68,000.00 (West)
- Growth Trend: +51.11% (North->West)

Generated Artifacts:
- outputs/extracted/sample_sales_table.csv
- outputs/visualizations/bar_chart_*.png
- outputs/summaries/sample_sales_summary.md

5. DUAL INTERFACES: INTERACTIVE TERMINAL WIZARD & NATIVE DESKTOP GUI

Provides both a rich terminal wizard and a zero-server Tkinter desktop window with live preview:



Interface Features:

- Task Selector: Instant switching between Summarize, VQA, Direct Qdrant Query, Visualization, CAD & Tables.
- Native Windows Explorer Picker: Graphical file selection with type filtering (*.pdf, *.png, *.csv, etc.).
- Threaded Non-Blocking Execution: Keeps GUI responsive while Ollama and Qdrant execute in background.
- Live Dual-Tab Results: Structured formatted markdown on Tab 1, and live visual preview/chart rendering on Tab 2.